NNUI_CELL

Addendum v0.3 — Workstream-A Realizations and the Discrete Interface-Tension Design

Supplement to the v0.2 Specification (which remains canonical)

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July 2026

Abstract

This addendum records what building workstream A — the agent-facing API, its integration demo, and the tessellation stage — forced into the discrete realization of NNUI_CELL. The continuum theory of v0.1/v0.2 is again unchanged. Five corrections (C5–C9) join the normative set: the discrete realization of E_{fill} with a wall-coverage stall predicate, plastic post-yield rest adaptation, bulk-transport binding, the latch's exact ξ=1 assignment, and radius conservation under lattice dilation. The final section is a *design*, not a result: the discrete realization of E_{cohesion} as boundary curvature flow, motivated by three measured failures of naive compactness mechanisms, with binary validation exits (T1–T4) that implementation must pass before the design is admitted as normative. Evidence levels are marked throughout: C5–C9 are validated (suite-green, screen-verified); the interface-tension design is internal reasoning awaiting its experiment.

1 Position of this document

The v0.2 specification remains the canonical statement of the model. Everything here concerns *discrete realization* — the same relationship v0.2's corrections C1–C4 bear to v0.1. Workstream A validated the API surface (milestones M1–M4: atomic grammar submission, the event-driven read path, LLM-driven layout, portable geometry-free cell records) and then confronted the layout semantics the prototype had deferred: the medium must actually fill its vessel, sizes must actually be solved from material state, and bound groups must actually cohere. Realizing those claims produced C5–C9 and exposed the one energy term still without a discrete form, E_{cohesion}.

2 Corrections C5–C9 (validated, normative)

C5. E_{fill} is a quasi-static drive with a wall-coverage stall. The fill phase advances matter outward at a rate that yields against solidity, $g = g_0\,(1 - S)$, entering the predictor as displacement (per C2). While filling, the frame is effectively accelerating (the boot pour): ξ rebuild is suspended per the §7.2 boundary condition, so the medium flows freely into void. Termination is *pressure stall*, not an area target — and the stall predicate may not fire while boundary spans remain unclaimed: edge-to-edge is part of E_{fill}'s definition ($\rho \to 1$ over Ω), not a preference. Measured: with an area-target stall the settled screen claimed 50–75% of the vessel; with the wall-coverage predicate, 95–100%. Near walls the drive direction blends toward the local void (matter creeps along walls; corner zones are owned by void-seeking); applied globally, void-seeking pulls cells into elongated shards — measured, rejected, and the restriction is therefore normative.

C6. Post-yield flow is plastic. While a layout region is awake (yielded), its material rest lengths follow realized geometry at a rate that yields against solidity, $\dot{r}_{\text{rest}} \propto (1 - S)$; at rest the rests hold, elastic below yield. This is the discrete realization of shear-thinning post-yield flow (n<1): committed matter gets out of the way and takes its new shape; stiff matter keeps its shape. Without plasticity the fill phase fights permanent elastic memory of the spawn lattice and the medium cannot tile.

C7. Binds transport bulk. The deliberate `bind` operation is interfacial: it must move *regions*, not points. A single-anchor spring fails in a plastic medium — the anchor migrates through the region while the bulk never moves (measured: bound chains left 170–240 px gaps). A stiffness-scaled attraction fails at rest — compliance stiffening turns it into a slingshot (the compliance-at-rest failure class, third sighting; measured eruption at 140 u/s). The normative form: a bounded, constant-scale quasi-static drive on *all* particles of both bound regions, active until the pair sits adjacent, with stiff contacts preventing overlap. Cohesive groups then pack, solved.

C8. The latch assigns $\xi = 1$ exactly. §7.3 writes "set $\xi_i = 1$" and means it. A latch that merely stops motion lets each region's ξ carry its private settle history — measured as identically-declared cells exporting different material state in their portable records (settle history leaking into the wire format). On latch engage, $\xi \leftarrow 1$: structure is fully rebuilt at rest, and exported material state is a function of declaration and physics, not of arrival order.

C9. Radius conservation under dilation. A dilated lattice is not a porous point-cloud: when lattice spacing grows, effective particle radius grows with it, $r_p = \frac{1}{2}\,\bar{\ell}$ (with $\bar{\ell}$ the region's mean lattice spacing), clamped to a sane range. Without this, regions finger through each other's interiors and settle interleaved (measured: hull-overlap "stacked sheets"). Two consequences are part of the correction: the C1 topological-guard tolerance must be *proportional* to the contact rest (an absolute tolerance misclassifies ordinary residual prestress at dilated scale), and contact-search structures must follow the radius. The material radius is exposed through the read path so a renderer draws true material extent — physical seams are seams, not voids.

Status: C5–C9 are validated — the engine suite (35+ tests including all seven invariants, the four milestone harnesses, frame budget, and API-boundary determinism) is green with all five active, and the settled screen is verified programmatically at 95–100% vessel claim.

3 Design: E_{cohesion} as boundary curvature flow

3.1 The residual defect and why it points here

With C5–C9 active, one defect class remains, measured and reproducible: *interleaving*. Arms of soft regions, stretched during early growth, freeze across neighbouring cells; per-region mean radius cannot seal non-uniformly stretched arms; vessel corners adjacent to such arms stay partially unclaimed. Nothing in the realized model penalizes elongation. In the continuum theory that penalty exists and has a name:

$$E_{\text{cohesion}} \;=\; w_c \sum_{(i,j)\in\partial} \sigma_{ij}\,\ell_{ij},$$

interface *length* minimization — surface tension. Foam tessellates because pressure fills while tension keeps every cell compact. We have built the pressure and not the tension.

3.2 Failed candidates (measured, excluded)

- *Global void-seeking* as a shape force: produces taffy shards — it is a fill term, not a tension term, and only survives restricted to the wall band (C5).
- *Radial roundness tension* (pull each particle toward the region's mean radius): isotropic, so it fights wall flattening and the fill drive; measured to collapse vessel claim to 70% and break solved sizes. The error is structural: it penalizes *radius variance*, but a wall-flattened tile has high radius variance and *low perimeter*. The energy says perimeter.

3.3 The design

Realize interface tension as **discrete curvature flow on region boundaries**:

1. **Boundary detection.** Per frame, a particle is a boundary particle of its region iff it has fewer than B same-region neighbours within radius ρ_n (existing spatial grid; index-order scan; deterministic).
2. **Discrete curvature.** For boundary particle i, let m_i be the mean position of its same-region *boundary* neighbours within ρ_b. The discrete curvature vector is

$$\kappa_i \;=\; m_i - x_i,$$

   the boundary Laplacian: on a straight (wall-flattened or shared-interface) edge $\kappa_i \approx 0$; on a convex bulge κ_i points inward; in a concave dent it points outward.
3. **Tension step.** Quasi-static displacement (C2), boundary particles of awake regions only:

$$\Delta x_i \;=\; \min\!\big(w_c' \,\|\kappa_i\|\,,\; c_{\max}\big)\,\hat{\kappa}_i\,\Delta t.$$

   Interior particles are untouched; the arrest phase and honest strain measurement are untouched; contacts and walls still project afterwards.

3.4 Why this succeeds where the candidates failed

Curvature flow minimizes perimeter directly: flat interfaces (shared seams, wall contact) carry no force, so it cannot fight wall conformity or tiling; bulges retract; dents fill; and a thin shard arm — whose boundary curvature is everywhere high and inward — retracts into the body of its cell. The predicted equilibrium is exactly the paper's stated consequence of E_{cohesion}: compact polygonal tiles, flat shared interfaces, and rounded free corners — coalescence and menisci. Because the force vanishes on flat boundaries it terminates naturally under the deadband and latch; no new stall machinery is required. It is diffusive and therefore stable at a bounded step ($c_{\max}$), and deterministic by construction.

3.5 Validation exits (binary, pre-registered)

Implementation is admitted as normative only if all four pass:

T1	*No interleaving.* For every reference layout, pairwise convex-hull intersection area between settled cells is $< 2\%$ of the smaller cell's area.
T2	*Corners claimed.* All four vessel corners pass the 50 px / 30% matter standard in every layout.
T3	*Nothing regresses.* Vessel claim stays $\geq 90\%$ lit; solved-size ratios, bind adjacency, phenotype expression, frozen-raster rest, and the §9 resize demonstration all hold.
T4	*The suite is the suite.* All engine tests green; bit-identical determinism preserved; the flicker-free sustained-rest window holds.

Status: design only — internal reasoning strength. T1–T4 are the experiment; the design is wrong if they fail, and the failure record joins this document's lineage the same way C1–C9 earned theirs.

4 Design revision D2: constraint-based tessellation

Added after implementation of the curvature-flow design. Status of the first pass, honestly: curvature flow itself behaved as designed (arms retract, compact cells) and survives; but the accompanying drives — fill, sibling exclusion, bind transport — were bounded competing displacements, and five exclusion formulations at multiple authorities all equilibrated *wherever their caps happened to balance* (pairwise hull overlap hovering near 4% against T1's 2% bar, with any strengthening breaking bind adjacency, solved sizes, or fill). The measured conclusion: **bounded competing drives cannot express the tessellation equilibrium.** Two genuine corrections were also forced and are kept: the *phase gate* (the fill phase is supra-yield by definition — no deadband arrest, no latching mid-pour; without it a stiff cell's sub-yield growth is reverted every frame and it latches at spawn size), and the *wall-wetting exemption* (tension acts on free surfaces; matter pressed against the vessel is wetting it, and tension at a corner tip evacuates the wedge E_{fill} claims).

4.1 The reformulation

Move the three competing terms *into* the XPBD projection loop, where equilibria are solved rather than raced:

1. **Fill as a per-region area constraint** (the paper's volume/occupancy constraint, §5.1). Per region r: $C_r = A_r(\mathbf{x}) - A_r^*$, with A_r the area of the region's ordered boundary polygon (convex hull at this fidelity) and targets solved from material state, $A_r^* \propto (1 - S_r)^2$ over the vessel area. The gradient on hull vertex v_i is the standard polygon-area gradient $\frac{1}{2}\,\text{perp}(v_{i+1} - v_{i-1})$; one scalar multiplier per region per frame. Because the multiplier accumulates inside the solve, elastic pullback cannot starve growth — the stuck-stiff-cell failure class dies structurally. Sizes remain *solved*: authority lives in the targets, not in growth-rate races.

2. **Exclusion largely for free.** Under hull-area constraints, wrapping is self-defeating: enclosed foreign space counts against the wrapper's own area budget while the enclosed cell's constraint pushes back from inside. Particle non-penetration (C1) plus conserved radii (C9) handle contact; an explicit exclusion survives only as a light correction for measured bites, if any remain.

3. **Bind as a plain XPBD distance constraint** (few anchor pairs, constant soft compliance — never stiffness-scaled, per C7). With area protected by its own constraint, a region can no longer comply by deforming around a migrating anchor: translation is the only compliant path, so binds transport bulk *because of* the constraint coupling, not despite it.

Curvature flow (§3), the phase gate, the wall-coverage stall (C5), plastic rests (C6), and radius conservation (C9) are unchanged. The same pre-registered exits T1–T4 gate this revision; T1 remains recorded-open until it passes with full fill.

5 New constants and dials

Symbol	Meaning	Status
g_0	fill drive scale (yields against S)	tuned (C5)
w_{band}	wall band for void-seeking	tuned (C5)
θ_{wall}	wall-coverage stall threshold ($> 90\%$)	tuned (C5)
p	plastic rest-adaptation rate ($\propto 1 - S$)	tuned (C6)
g_b	bind bulk-drive scale (bounded, constant)	tuned (C7)
r_p	effective particle radius, $\frac{1}{2}\bar{\ell}$	derived (C9)
B, ρ_n, ρ_b	boundary detection neighbourhood	design (T1–T4)
$w'_c, c_{\max}$	curvature-flow weight and step clamp	design (T1–T4)

As with the v0.2 operating point, the corrections are qualitative and forced by measured failures; the constants are tuned, not derived.

MIVIM Development — Standards in Novel Design. Addendum v0.3, July 2026; supplements the v0.2 specification, which remains canonical. All results referenced herein are reproducible from the repository sources (`engine/`, `demo/`) and their automated gates.